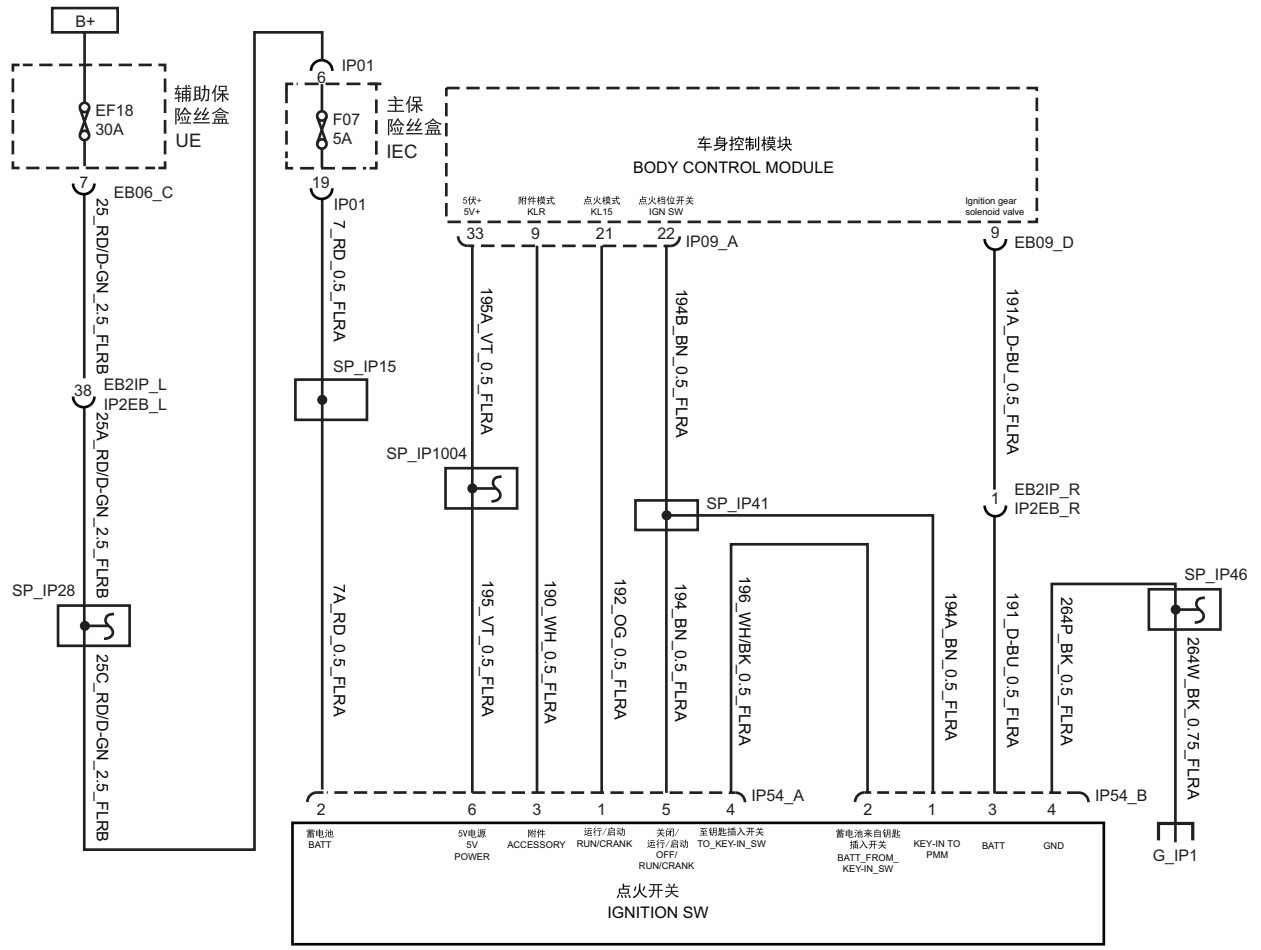
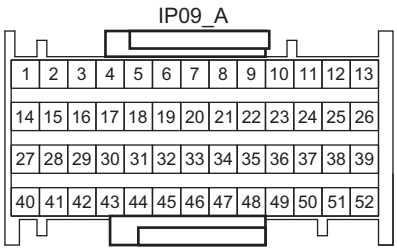
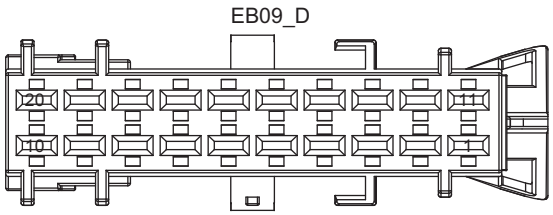
B1600-11、 B1600-12、 B1600-13、 B1600-1E

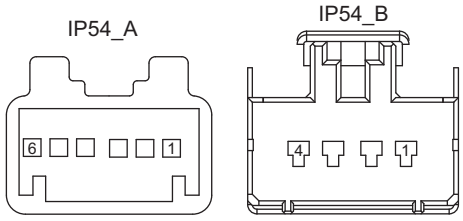
Fault diagnosis code
B1600-11：Ignition gear switch circuit short to ground
B1600-12：Ignition gear switch circuit short to battery
B1600-13：Ignition gear switch cricuit open
B1600-1E：Ignition gear switch circuit resistance out of range
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM module fault • Ignition switch module fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1600-11
Ignition gear switch circuit short to ground
B1600-12
Ignition gear switch circuit short to battery
B1600-13
Ignition gear switch circuit open
B1600-1E
The input resistance of front ignition gear switch is detected to be out of range for more than 1s.
To set the effect of a fault code condition
Ignition gear switch function failure
Description of circuit diagram

Circuit diagram

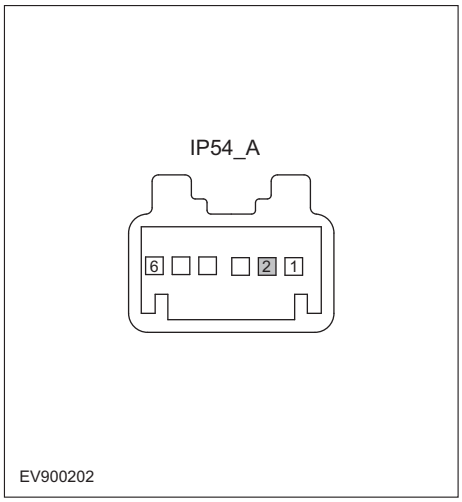


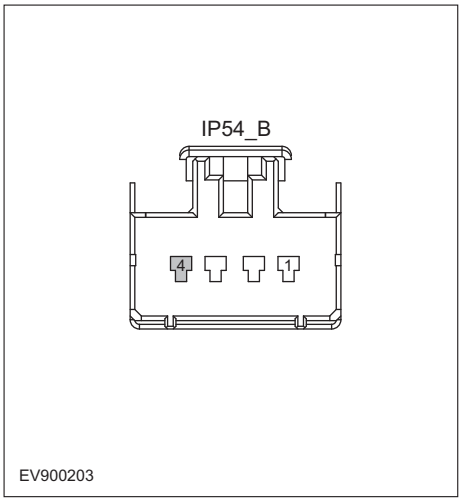




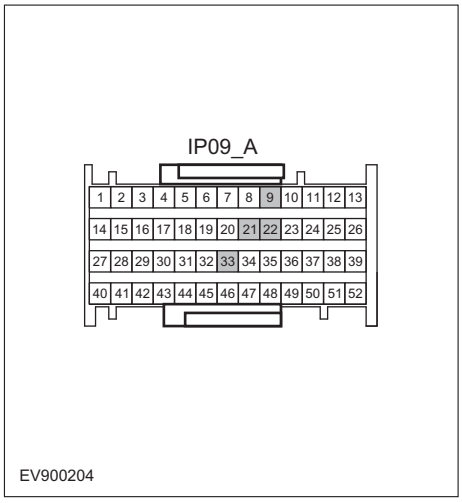


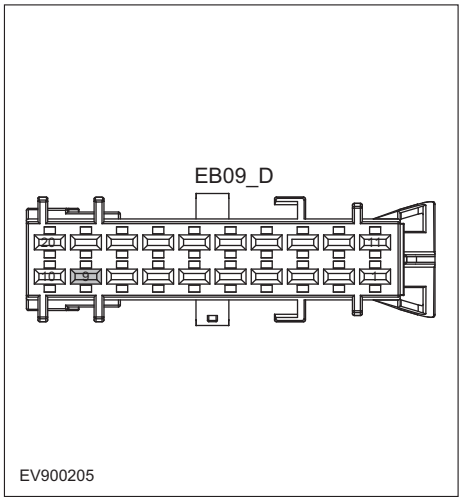
EV900201

<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check fuses EF18/F07 for damage.
2 Check battery capacity.
1 Check the harness plug of ignition switch moudle, BCM for damage, poor contact, aging and looseness.
2 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Diagnose the BCM with diagnostic apparatus.
4 Is there any other fault code except for B1600-11、B1600-12、 B1600-13、 B1600-1E?</td></tr>
<tr><td colspan="2"></td><td>YES ➞ Refer to:DTC Summary list(BCM).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the power voltage of ignition switch battery.</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug IP54_A, IP54_B of ignition switch.
3 Measure the voltage between the harness terminal of ignition switch and ground.
Measuring circuit: voltage between terminal 2 on plug IP54_A and ground.
Standard value:10~14V
4 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Test whether the grounding circuit of ignition switch is open.
	1 Measure the resistance between the harness terminal of ignition switch and ground. Measuring circuit: resistance between terminal 4 on plug IP54_B and ground. **Standard value: < 1Ω** 2 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Test whether the circuit of ignition switch of BCM is open.





1 Connect the harness plug IP54_A, IP54_B of ignition switch.

2 Turn the ignition switch to OFF.

3 Disconnect the harness plug IP09_A, EB09_D of BCM module.

4 Measure the resistance between the harness terminal of BCM and ground.

Measuring circuit: resistance between terminal 9 on plug IP09_A and ground.

Measuring circuit: resistance between terminal 21 on plug IP09_A and ground.

Measuring circuit: resistance between terminal 22 on plug IP09_A and ground.

Measuring circuit: resistance between terminal 33 on plug IP09_A and ground.

Measuring circuit: resistance between terminal 9 on plug EB09_D and ground.

Standard value: ∞

5 Measure the voltage between the harness terminal of BCM and ground.

Measuring circuit: voltage between terminal 9 on plug IP09_A and ground.

Measuring circuit: voltage between terminal 21 on plug IP09_A and ground.

Measuring circuit: voltage between terminal 22 on plug IP09_A and ground.

Measuring circuit: voltage between terminal 33 on plug IP09_A and ground.

Measuring circuit: voltage between terminal 9 on plug EB09_D and ground.

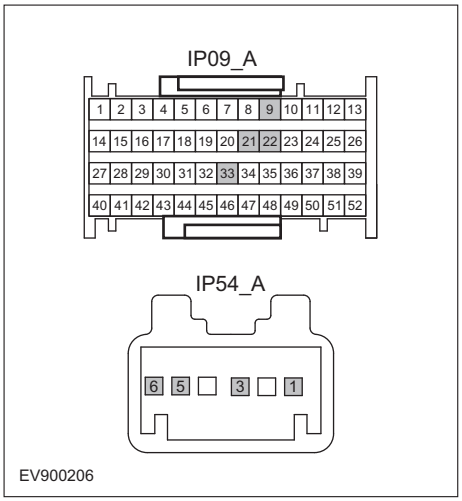
Standard value: ≈ 0V

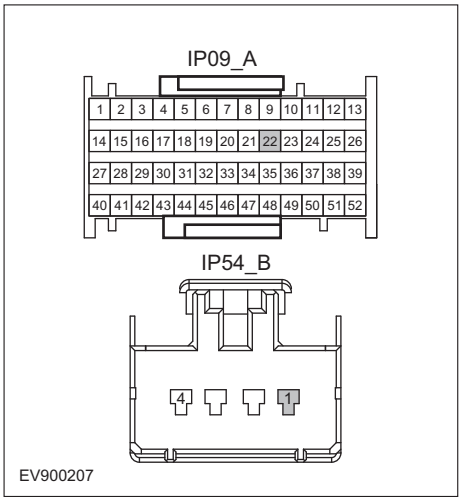
6 Check whether the result is normal or not?

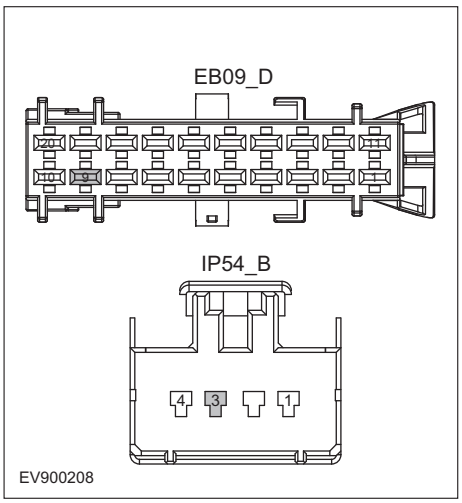
NO → Repair or replace the faulted parts.

YES ↓

Step 6	Check the circuit between ignition switch and BCM (open circuit).







1 Turn the ignition switch to OFF.

2 Disconnect the harness plug IP54_A, IP54_B of ignition switch.

3 Measure the resistance between the harness terminal of ignition switch and the BCM.

Measuring circuit: resistance between terminal 1 on plug IP54_A and terminal 21 on plug IP09_A.

Measuring circuit: resistance between terminal 3 on plug IP54_A and terminal 9 on plug IP09_A.

Measuring circuit: resistance between terminal 5 on plug IP54_A and terminal 22 on plug IP09_A.

Measuring circuit: resistance between terminal 6 on plug IP54_A and terminal 33 on plug IP09_A.

Measuring circuit: resistance between terminal 1 on plug IP54_B and terminal 22 on plug IP09_A.

Measuring circuit: resistance between terminal 3 on plug IP54_B and terminal 9 on plug EB09_D.

Standard value: ≈ 0Ω

4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 7	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the BCM Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	